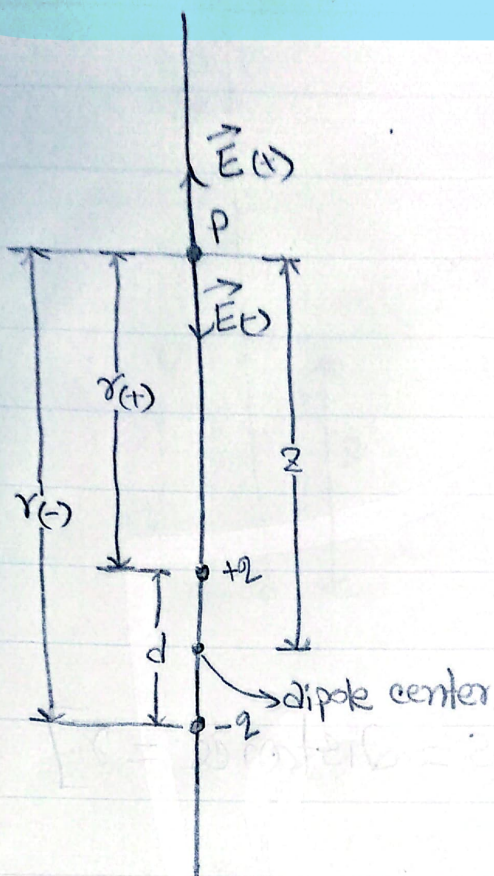


①



Date

Electric Field Due to an Electric Dipole.



Here,

$$r(+) = z - \frac{d}{2}$$

$$r(-) = z + \frac{d}{2}$$

Electric field at point P,

$$E = E(+) - E(-)$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q}{r(+)^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{r(-)^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q}{(z - \frac{d}{2})^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{(z + \frac{d}{2})^2}$$

$$\therefore E = \frac{1}{2\pi\epsilon_0} \frac{qd}{z^3}$$

$$\therefore E = \frac{1}{2\pi\epsilon_0} \frac{p}{z^3}$$

$$p = qd$$



Potential Due to a Charge Particle

we know,

$$dW = \vec{F} \cdot d\vec{s}$$

$$\Rightarrow dW = q_0 \vec{E} \cdot d\vec{s}$$

$$\Rightarrow W = \int_i^f q_0 \vec{E} \cdot d\vec{s}$$

$$\Rightarrow \frac{W}{q_0} = \int_i^f \vec{E} \cdot d\vec{r} \quad [s = \text{distance} = r]$$

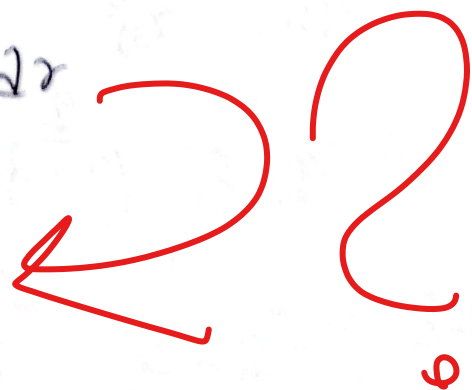
$$\Rightarrow V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{r}$$

$$\Rightarrow 0 - V = - \int_R^\infty E dr$$

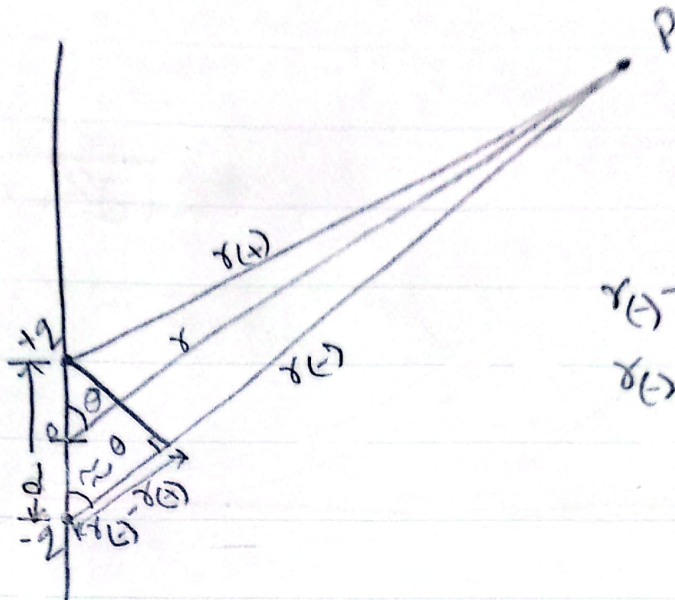
$$\Rightarrow V = \frac{q}{4\pi\epsilon_0} \int_R^\infty \frac{1}{r^2} dr$$

$$= \frac{q}{4\pi\epsilon_0} \left[-\frac{1}{r} \right]_R^\infty$$

$$\therefore V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$



Potential Due to an Electric Dipole



$$r(-) - r(+)= d \cos \theta$$

$$r(-) \cdot r(+)= r^2$$

$$V = V_{(+)} + V_{(-)}$$

$$\Rightarrow V = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r(+)} - \frac{q}{r(-)} \right)$$

$$\Rightarrow V = \frac{q}{4\pi\epsilon_0} \frac{r(-) - r(+)}{r(-)r(+)}$$

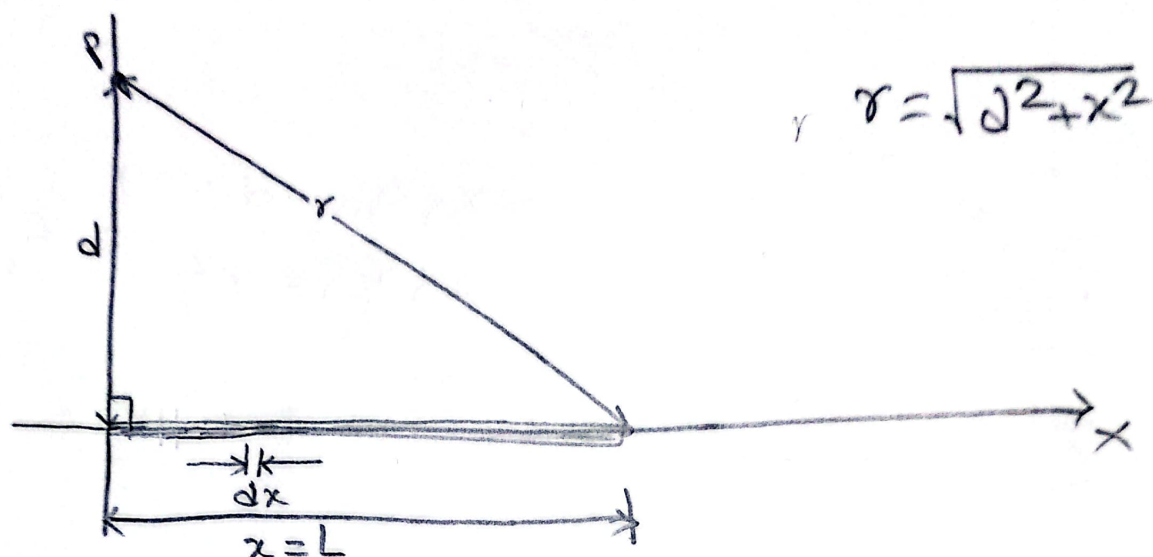
$$\Rightarrow V = \frac{1}{4\pi\epsilon_0} \frac{qd \cos \theta}{r^2}$$

$$\therefore V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$$

~~*~~

(4)

Potential Due to A Continuous Charge Distribution



$\lambda = \text{charge density}$

$$\therefore dq = \lambda dx$$

$$\therefore dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{(x^2 + d^2)^{3/2}}$$

$$\therefore V = \int dV = \int_0^L \frac{1}{4\pi\epsilon_0} \frac{\lambda}{(x^2 + d^2)^{3/2}} dx$$

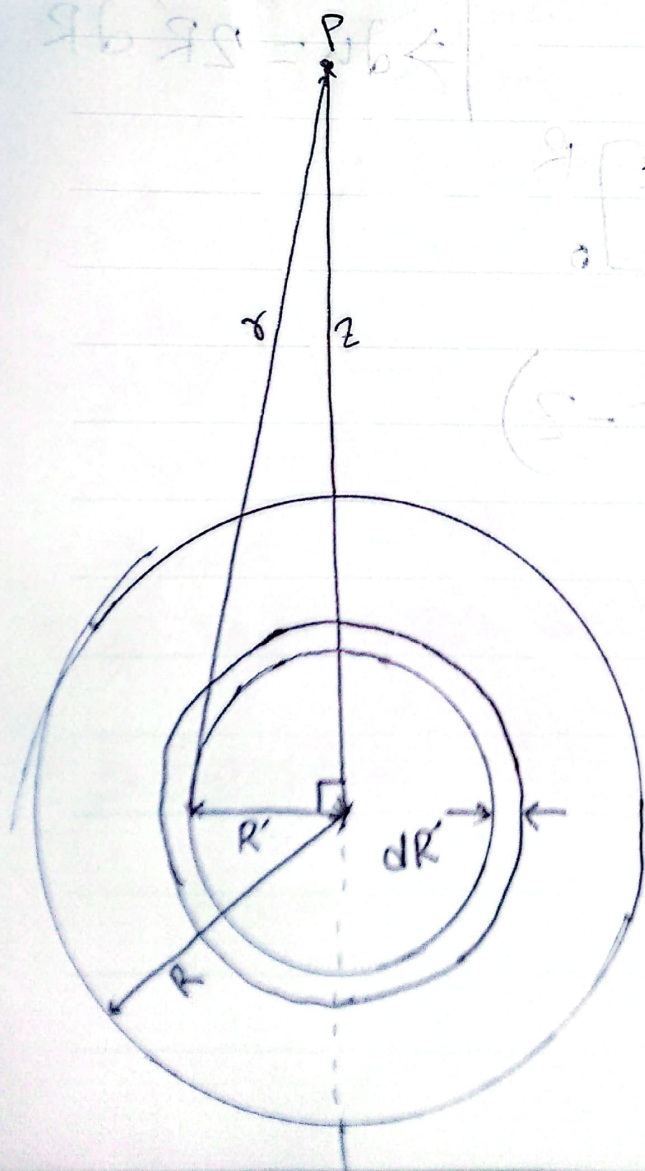
$$\Rightarrow \boxed{V} = \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{dx}{(x^2 + d^2)^{3/2}}$$

$$\Rightarrow V = \frac{\lambda}{4\pi\epsilon_0} \left[\ln(x + (x^2 + d^2)^{1/2}) \right]_0^L$$

$$\Rightarrow V = \frac{\lambda}{4\pi\epsilon_0} \left[\ln(L + (L^2 + d^2)^{1/2}) - \ln d \right]$$

$$\therefore V = \frac{\lambda}{4\pi\epsilon_0} \ln \left(\frac{L + \sqrt{L^2 + d^2}}{d} \right)$$

~~★~~ # Potential due to a Charge Disk.



$$q = \sigma A$$

$$\Rightarrow \frac{dq}{dr} = \sigma \frac{dA}{dr}$$

$$\Rightarrow \frac{dq}{dr} = \sigma \frac{d(\pi r^2)}{dr}$$

$$\Rightarrow \frac{dq}{dr} = \sigma 2\pi r$$

$$\therefore dq = \sigma 2\pi R' dR' [r = R']$$

$$\text{Here, } r = \sqrt{R'^2 + z^2}$$

$$dV = \frac{1}{4\pi\epsilon_0} \frac{\sigma 2\pi R' dR'}{\sqrt{z^2 + R'^2}}$$

(6)

$$\therefore V = \int dV = \int_0^R \frac{1}{4\pi\epsilon_0} \frac{\sigma 2\pi R' dR'}{\sqrt{z^2 + R'^2}}$$

$$\Rightarrow V = \frac{\sigma \phi \pi}{4\pi\epsilon_0} \int_0^R \frac{2R'}{\sqrt{z^2 + R'^2}} dR'$$

$$\Rightarrow V = \frac{\sigma}{4\epsilon_0} \int_0^R \frac{du}{\sqrt{u}}$$

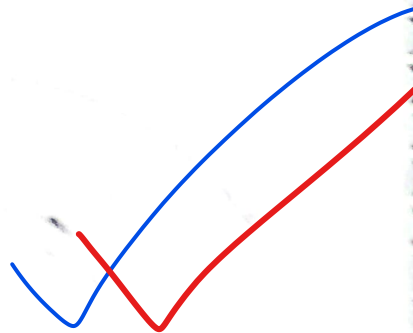
$$= \frac{\sigma}{4\epsilon_0} \left[\frac{u^{1/2}}{1/2} \right]_0^R$$

$$= \frac{\sigma}{4\epsilon_0} \left[\frac{u^{1/2}}{1/2} \right]_0^R$$

$$= \frac{\sigma}{2\epsilon_0} \left[\sqrt{z^2 + R'^2} \right]_0^R$$

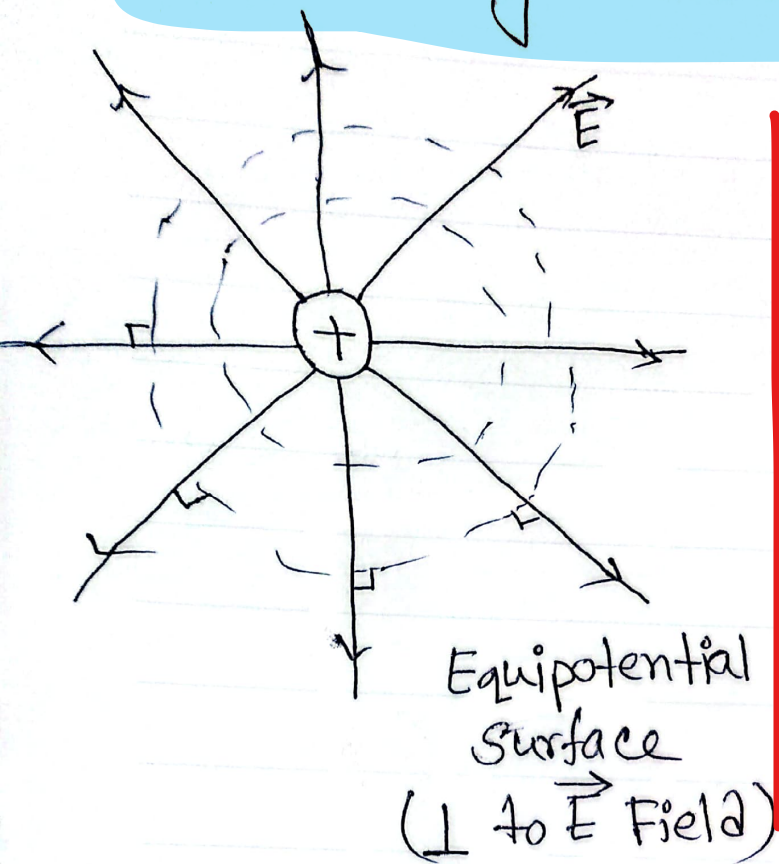
Let, $u = z^2 + R'^2$
 $\Rightarrow du = 2R' dR'$

$$\therefore V = \frac{\sigma}{2\epsilon_0} (\sqrt{z^2 + R^2} - z)$$



End

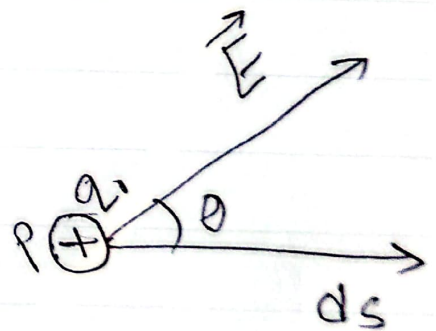
Calculating Field from Potential.



$$\begin{aligned}
 V &= \frac{1}{4\pi\epsilon_0} \frac{q}{r} \\
 \Rightarrow \frac{dV}{dr} &= \frac{d}{dr} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r} \right) \\
 &= \frac{q}{4\pi\epsilon_0} \frac{d}{dr} (r^{-1}) \\
 &= -\frac{q}{4\pi\epsilon_0} \frac{1}{r^2} \\
 \therefore E &= -\frac{dV}{dr}
 \end{aligned}$$

Alternative

$$\begin{aligned}
 V &= -\frac{w}{q} \\
 \Rightarrow dV &= -\frac{dw}{q} \\
 \Rightarrow -q dV &= dw = q_0 E \cos\theta ds \\
 \Rightarrow E \cos\theta &= -\frac{dV}{ds} \\
 \therefore E ds &= -\frac{dV}{\cos\theta}
 \end{aligned}$$



Link